Conceptual Note: Multi-Agent Delivery System in Crowded Environment

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1 Agent Conceptualisation

Colruyt recently launched a pilot program in Leuven using self-driving vehicles for grocery delivery. Operating in a busy city centre, these vehicles must navigate unstructured pedestrian traffic, providing the real-world inspiration for the multi-agent simulation developed in this study.

The agent under study is an autonomous delivery robot operating within a structured urban environment analogous to a city centre. Upon receiving a delivery task, the agent processes environmental information and derives a route according to its internal decision logic. We formalise the problem as an instance of Multi-Agent Pickup and Delivery (MAPD) (Xu et al., 2022), in which a set of k agents $\mathcal{A} = \{a_1, a_2, \ldots, a_k\}$ execute incoming tasks defined by pairs of pickup and delivery locations on an undirected connected graph $G = (V, E)$.

We assume that the agent maintains partial observability over three categories of information: (i) the distances between its current position, the pickup location, and the designated delivery node; (ii) the spatial positions of other active robots in the fleet; and (iii) an estimate of pedestrian congestion for each traversable zone. Based on these inputs, the agent selects a route that optimises its primary objective function. Pedestrians constitute what Bonalumi et al. (2025) term *external agents*: moving entities whose goals and planned paths are unknown to the delivery robots, and who do not yield to avoid collisions. This asymmetric interaction model is the defining characteristic of the MAPD with External Agents (MAPD-EA) formulation, which most closely captures the structure of our simulation environment.

2 The Phenomenon of Interest: Congestion Delivery Time

Delivery time is a central performance metric in last-mile logistics and the primary dependent variable in this study. We adopt the standard MAPD *makespan* — the time elapsed from the release of the first task to the completion of the last task (Lodigiani, 2021; Xu et al., 2022). Robot behaviour is expected to vary with the quality and granularity of the environmental information available to the agent. Specifically, this study investigates the main hypothesis (H1) and related sub-hypotheses (H2–H4):

H1 Agents with real-time congestion information and information about other agents' locations will have lower delivery times than agents operating without such information. This advantage will increase as traffic congestion rises.

- **H2** Under high-congestion conditions, agents will avoid dense zones and prefer longer but less obstructed routes.
- **H3** There is a congestion threshold above which congestion-aware routing yields statistically significant reductions in total delivery time relative to shortest-path routing.
- **H4** Knowledge of other robots' current positions will improve overall delivery time, independently of the traffic congestion level.

3 Agent Behaviour

The robots' observable behavior is described by two key metrics: the spatial trajectory followed during task execution and the total makespan, defined as the interval between task assignment and successful delivery. We consider three routing strategies—shortest-path routing (SSP routing), environment-aware routing (EV routing), and a strategy that incorporates both environmental and agent awareness (EA strategy) —each reflecting a different balance between path length, congestion exposure, and sensitivity to other agents.

The task assignment component of each strategy follows a greedy nearest-task heuristic, in which a free agent selects the unassigned task whose pickup location is closest to its current position. Xu et al. (2022) demonstrate that assigning a full sequence of tasks to each agent — rather than one task at a time — consistently reduces makespan; this motivates the use of pre-planned routes in our experimental design.

We hypothesise that the degree of behavioural divergence between the SSP routing and the EV routing will increase monotonically with pedestrian density, while the EA strategy will consistently outperform the other two strategies. Under low-density conditions the EV routing and SSP routing are expected to converge on near-equivalent performance. Bonalumi et al. (2025) provide direct empirical support for this hypothesis: their Markovian external-agent model reduces conflicts by up to 85% relative to a purely reactive baseline in constrained corridor environments, while the benefit is more modest in unconstrained open spaces. The magnitude of the density–performance interaction constitutes the primary quantity of interest in the present study.

4 Experimental Setup

The study adopts an offline planning paradigm in which all pickup and delivery locations are known prior to route execution. Agents plan their complete trajectories before departure and do not revise these plans in response to environmental changes encountered during traversal. This design choice isolates the effect of pre-departure information quality from the confounding influence of online replanning capabilities, enabling a controlled assessment of each routing strategy.

The simulation environment, illustrated in Figure 1, consists of a circular road network modelling a simplified urban centre, populated with a variable number of pedestrian agents. Pedestrian density is manipulated as the primary independent variable and routing strategy as the secondary independent variable. Performance is evaluated over a fixed set of delivery tasks, with makespan as the dependent variable.

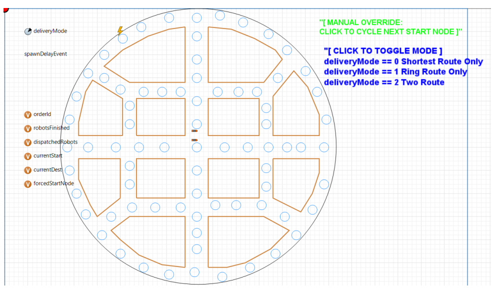

Figure 1: Anylogic Environment Multi-agent Delivery System

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